

4(a)	1 temperature increases and milk yield decreases ; 2 comparative data quote – April = 10 °C and 792–798 kg cow⁻¹, August = 18 °C and 711–716 kg cow⁻¹ ;	2
4(b)	1 continuous and (milk yield) does not fall into distinct classes / shows a range / intermediates / no distinct categories / AW ; 2 environment and (milk yield) affected by air temperature ; 3 genetic / polygenic and many / 20, genes have effect (on milk yield) or heat-tolerant cattle have lower milk yield than Holstein Friesian cattle ;	3

Question	Answer	Marks									
4(c)	any three from: 1 SLICK allele increases milk yield (in both months / at both temperatures) ; 2 in March / at 5 °C, there is a small difference (between milk yield) ; 3 in September / at 14 °C, there is a larger difference (between milk yields) ; 4 cattle with SLICK allele maintain milk yield (in both months / at both temperatures) ; 5 comparative data quote to support mp1–mp4 ; <table border="1"> <tr> <th>Month</th><th>Holstein without SLICK / kg cow⁻¹</th><th>Holstein with SLICK / kg cow⁻¹</th></tr> <tr> <td>March / at 5 °C</td><td>780–785</td><td>795–800</td></tr> <tr> <td>Sept / at 14 °C</td><td>690–700</td><td>770–780</td></tr> </table>	Month	Holstein without SLICK / kg cow ⁻¹	Holstein with SLICK / kg cow ⁻¹	March / at 5 °C	780–785	795–800	Sept / at 14 °C	690–700	770–780	3
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